1 Solution — GIAM 7.1 Problem 10

1.1 Problem

How many functions are there from a set of size n to a set of size m?

1.2 Setup

Let A be a set with $|A| = n$ and B a set with $|B| = m$ (both $n, m \geq 0$). A *function* $f : A \to B$ assigns to each $a \in A$ a single image $f(a) \in B$; two functions agree iff they take the same value at every $a \in A$. Denote the set of all such functions by

$$B^A \;:=\; \{\, f \;:\; A \to B \,\}.$$

We wish to compute $|B^A|$.

1.3 Claim

$$\boxed{|B^A| \;=\; m^n.}$$

We give two proofs.

1.4 Proof 1 — Element-by-Element Choice

Specifying $f : A \to B$ amounts to picking, *independently for each* $a \in A$, a single image $f(a) \in B$. With n elements of A and m choices per element, the multiplication principle gives

$$|B^A| \;=\; \underbrace{m \cdot m \cdots m}_{n \text{ factors}} \;=\; m^n.$$

More formally, enumerate $A = \{a_1, a_2, \ldots, a_n\}$ and consider the *tuple* map

$$\Phi \;:\; B^A \;\longrightarrow\; B^n, \qquad \Phi(f) \;=\; \big(f(a_1),\, f(a_2),\, \ldots,\, f(a_n)\big).$$

This Φ is a bijection: any tuple $(b_1, \ldots, b_n) \in B^n$ pulls back to the unique function defined by $f(a_i) = b_i$. Hence

$$|B^A| \;=\; |B^n| \;=\; m^n. \qquad \blacksquare$$

1.5 Proof 2 — Induction on n

Base case $(n = 0)$. Then $A = \emptyset$. There is exactly one function from $\emptyset$ to B — the *empty function*, whose graph is the empty set — vacuously satisfying the definition. So $|B^{\emptyset}| = 1 = m^0$. ✓

Inductive step. Suppose $|B^{A'}| = m^n$ for every set A' with $|A'| = n$. Let $|A| = n + 1$; fix any $a \in A$ and write $A' = A \setminus \{a\}$, so $|A'| = n$. Specifying $f : A \to B$ is equivalent to specifying the two independent data

- its restriction $f|_{A'} : A' \to B$, of which there are m^n by the inductive hypothesis;
- the value $f(a) \in B$, of which there are m.

By the multiplication principle, $|B^A| = m^n \cdot m = m^{n+1}$. ✓

By induction, $|B^A| = m^{|A|}$ for every finite A. $\blacksquare$

1.6 Sanity Check on Small (n, m)

n	m	m^n	**Comment**
0	5	1	The empty function $\emptyset \to B$ is unique.
3	0	0	No function from a nonempty set to $\emptyset$ (nowhere to send the elements).
1	3	3	A function $\{a\} \to \{1, 2, 3\}$ is determined by $f(a)$: three choices.
2	2	4	$(f(a), f(b)) \in \{0, 1\}^2$ — the four pairs $(0, 0), (0, 1), (1, 0), (1, 1)$.
3	2	8	Eight binary triples — equivalently, the eight subsets of $\{a, b, c\}$.

Table 1.1

Note the corner case $0^0 = 1$: there is one function $\emptyset \to \emptyset$, namely the empty function. This is the convention required to make the formula uniform.

1.7 Connection — The Special Case $m = 2$ Recovers Subsets

Set $B = \{0, 1\}$. A function $f : A \to \{0, 1\}$ is the *characteristic function* of the subset

$$A_f \;=\; \{\, a \in A \;:\; f(a) = 1 \,\} \;\subseteq\; A,$$

and conversely every subset $S \subseteq A$ defines a characteristic function

$$\chi_S(a) \;=\; \begin{cases} 1 & \text{if } a \in S, \\ 0 & \text{otherwise.} \end{cases}$$

The two assignments are mutually inverse bijections between $\{0,1\}^A$ and $\mathcal{P}(A)$, so

$$|\mathcal{P}(A)| \;=\; |\{0,1\}^A| \;=\; 2^n,$$

which is exactly Problem 8. In this light Problem 8 is the special case $m = 2$ of the present problem.

1.8 Remark — Notation B^A

The notation B^A for the set of functions $A \to B$ is *chosen* so that

$$|B^A| \;=\; |B|^{|A|}.$$

The exponent style is more than a mnemonic: it is consistent with cardinal arithmetic for infinite sets, where Cantor's theorem (§8.3) gives $2^{|A|} > |A|$ — the infinite-set generalization of the inequality $2^n > n$.

1.9 Remark — Other Standard Counts

For completeness — and as a guide for future problems — the closely related counts of *restricted* functions take different forms. Let $|A| = n$ and $|B| = m$.

Type of function $f : A \to B$	**Count**

All functions	m^n (this problem)
Injective ($f(a) = f(a') \Rightarrow a = a'$)	$\dfrac{m!}{(m-n)!} = m(m-1)\cdots(m-n+1)$ (requires $m \geq n$)
Bijective	$n!$ if $m = n$, else 0
Surjective	$\sum_{k=0}^{m}(-1)^k\binom{m}{k}(m-k)^n$ (inclusion–exclusion)

Table 1.2

Only the *all-functions* count is asked for here; the others rely on additional ideas (subset choice with order, factorials, inclusion–exclusion) developed later in the chapter.